STABILIZATION OF THE DIMENSION VECTORS BY THE STABILITY PARAMETER: THE FACE STRUCTURE OF A ZETA-BULLET-STABLE MODULE

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Allowed standard prerequisites (editorial).

- The stability parameter zeta-bullet, its faces in the space of stability conditions, and the wall-and-chamber structure attached to the dimension vectors. Developed in the author's subsection on stability (source lines 726-948) and used throughout the counted proof. Supplies the face decomposition with respect to which the module is analysed and the walls along which the dimension vectors stabilize.
- The monotonicity result for the dimension vectors along a wall, attributed by the author to the cited paper Na-quiver. Used inside the counted proof and attributed by the author to section 5 of Na-quiver. The monotonicity statement is what makes the analysis pass from a general point of the face to the extremal ones.
- The affine-quiverPlumbing criterion: the author's earlier theorem on which dimension vectors admit a non-empty regular part, used at the end of the argument. The author's own statement at source lines 853-898 (label thm:Rudakov). Identifies the case in which the stabilization claim of the lemma is sharp.

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Source reference. Original lemma 2.7: sources/algebra-fl01-quiver-varieties-and-branching/pinned-original-source.tex, source lines 968–968. Navigation pointer only: the counted proof appeals to this lemma at one place in the induction and the property it supplies is declared as a named standard prerequisite; the lemma's own statement and proof are not reproduced here.

Source reference. Original lemma 2.2: sources/algebra-fl01-quiver-varieties-and-branching/pinned-original-source.tex, source lines 792–792. Navigation pointer only: it is the author's slope comparison lemma, which the counted proof recalls in one step of the stability analysis; the lemma itself is not reproduced here.

Source reference. Original theorem 2.4: sources/algebra-fl01-quiver-varieties-and-branching/pinned-original-source.tex, source lines 853–853. Navigation pointer only: the counted proof uses the author's earlier criterion for the non-vanishing of the regular part at the end of the argument, and its content is declared as a named standard prerequisite rather than reproduced.

Lemma 0.1. *Suppose $W \neq 0$.*

(1) (cf. [4, 2.8]) *Suppose ζ is in a chamber. Then ζ-semistability implies ζ-stability, and hence $\mathfrak{M}_\zeta^{\mathrm{s}}(\mathbf{v},\mathbf{w}) = \mathfrak{M}_\zeta(\mathbf{v},\mathbf{w})$.*

(2) (cf. [11, 1.4]) *If two stability parameters ζ, ζ' are contained in the same face F, ζ-stability (resp. ζ-semistability) is equivalent to ζ'-stability (resp. ζ'-semistability).*

(3) *Suppose stability parameters ζ, $\zeta^\bullet$ are contained in F and $F^\bullet$ respectively. If $F^\bullet \subset \overline{F}$, then*

(a) *a ζ-semistable module is $\zeta^\bullet$-semistable,*
(b) *a $\zeta^\bullet$-stable module is ζ-stable.*

Proof. (1) For given V,W, we define $\tilde{\zeta} = (\zeta, \zeta_\infty)$ with $\theta_{\tilde{\zeta}}(V,W) = 0$ as before. Suppose that (B,a,b) is $\tilde{\zeta}$-semistable, but not $\tilde{\zeta}$-stable. We take a Jordan–Hölder filtration as in Theorem 2.4(2). Since $\delta_k W/\delta_{k+1}W \neq 0$ only for one k in $0,\dots,N$, we have a k with $\delta_k W/\delta_{k+1}W = 0$ from the assumption $N \geq 1$. As the quotient module $\mathrm{gr}_k(B,a,b) = (B,a,b)|_{(V^k,\delta_k W)/(V^{k+1},\delta_{k+1}W)}$ is $\tilde{\zeta}$-stable, its dimension vector $\dim V^k - \dim V^{k+1}$ (with the ∞-component is 0) is in $R_+(\mathbf{v})$ by Lemma 2.7. Moreover, by the condition in the Jordan–Hölder filtration, we have

$$\theta_{\tilde{\zeta}}(\mathrm{gr}_k(B,a,b)) = \theta_{\tilde{\zeta}}(V,W).$$

The right hand side is equal to 0 by our convention, and the left hand side is equal to $\zeta\cdot(\dim V^k - \dim V^{k+1})$ up to scalar. This contradicts with our assumption that ζ is in a chamber.

(2) We define $\tilde{\zeta}$, $\tilde{\zeta}'$ as before. Suppose that $(B,a,b) \in \mu^{-1}(0) \subset \mathbf{M}(\mathbf{v},\mathbf{w})$ is $\tilde{\zeta}$-semistable and is not $\tilde{\zeta}'$-semistable. We take the Harder–Narasimhan filtration for (B,a,b) with respect to the $\tilde{\zeta}'$-stability as in Theorem 2.4(1). We have $N \geq 1$ from the assumption.

Consider first the case when there exists $0 \leq k \leq k_W - 1$ with $\theta_{\tilde{\zeta}'}(\mathrm{gr}_k(B,a,b)) < 0$. Then $\theta_{\tilde{\zeta}'}(\mathrm{gr}_l(B,a,b)) < 0$ for any $0 \leq l \leq k$. On the other hand from the ζ-semistability of (B,a,b), we have $\theta_{\tilde{\zeta}}((V,W)/(V^k,\delta_k W)) \geq 0$ by Lemma 2.2. Therefore there exists at least one l in $[0,k]$ with $\theta_{\tilde{\zeta}}(\mathrm{gr}_l(B,a,b)) \geq 0$. We further take a Jordan–Hölder filtration of $\mathrm{gr}_l(B,a,b)$ to find $\tilde{\zeta}'$-stable representations V' with ${}^{\zeta'\cdot\dim V'}/_{\sum \dim V_i'} = \theta_{\tilde{\zeta}'}(V',0) = \theta_{\tilde{\zeta}'}(\mathrm{gr}_l(B,a,b)) < 0$. Note that $\mathrm{gr}_l(B,a,b)$, and hence V' has 0 in the W-component. Since $\theta_{\tilde{\zeta}}(\mathrm{gr}_l(B,a,b)) \geq 0$, we can take V' so that ${}^{\zeta\cdot\dim V'}/_{\sum \dim V_i'} = \theta_{\tilde{\zeta}}(V',0) \geq 0$. As $\dim V' \in R_+(\mathbf{v})$, this contradicts with the assumption that ζ and ζ' are in the common face.

The same argument leads to a contradiction in the case when there exists $k_W + 1 \leq k \leq N$ with $\theta_{\tilde{\zeta}'}(\mathrm{gr}_k(B,a,b)) > 0$. Since $N \geq 1$ and $\theta_{\tilde{\zeta}'}(V,W) = 0$, at least one of two cases actually occur. Therefore (B,a,b) is ζ'-semistable.

Suppose further that (B,a,b) is ζ-stable. We want to show that it is also ζ'-stable. Assume not, and take a Jordan–Hölder filtration of (B,a,b) with respect to the ζ'-stability. Either of $\mathrm{gr}_0(B,a,b)$ and $\mathrm{gr}_N(B,a,b)$ has the W-component 0. Suppose the first one has the W-component 0. We have ${}^{\zeta' \cdot \dim(V/V^1)}/_{\sum \dim V_i/V_i^1} = \theta_{\tilde{\zeta}'}(\mathrm{gr}_0(B,a,b)) = 0$. On the other hand, the ζ-stability of (B,a,b) implies ${}^{\zeta \cdot \dim(V/V^1)}/_{\sum \dim V_i/V_i^1} = \theta_{\tilde{\zeta}}(\mathrm{gr}_0(B,a,b)) > 0$. This contradicts with the assumption. The same argument applies to the case when $\mathrm{gr}_N(B,a,b)$ has the W-component 0. Therefore (B,a,b) is ζ'-stable.

(3) Take a sequence $\{\zeta_n\}$ in F converging to $\zeta^\bullet$. Take a nonzero submodule $(S,\delta W)$ of (B,a,b). (a) From the ζ-semistability and (2), we have $\theta_{\tilde{\zeta}_n}(S,\delta W) \leq \theta_{\tilde{\zeta}_n}(V,W)$ for any n. Taking limit, we get $\theta_{\tilde{\zeta}^\bullet}(S,\delta W) \leq \theta_{\tilde{\zeta}^\bullet}(V,W)$. Therefore (B,a,b) is $\zeta^\bullet$-semistable. (b) We assume $(S,\delta W) \neq (V,W)$. From the $\zeta^\bullet$-stability, we have $\theta_{\tilde{\zeta}^\bullet}(S,\delta W) < \theta_{\tilde{\zeta}^\bullet}(V,W)$. Then $\theta_{\tilde{\zeta}_n}(S,\delta W) < \theta_{\tilde{\zeta}_n}(V,W)$ for sufficiently large n. Therefore (B,a,b) is ζ_n-stable. By (2) it is also ζ-stable. □

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